1 Workshop Outline

i. Go through LaTeX guide on website, and homework templates.

   (a) Our 22 Guide.
   (b) Our Symbol Guide.

ii. Enumerations

   i. $x + 1 = 4$
   ii. $x = 3$
   iii. night

   1. hi
   2. hello

iii. Math Expressions

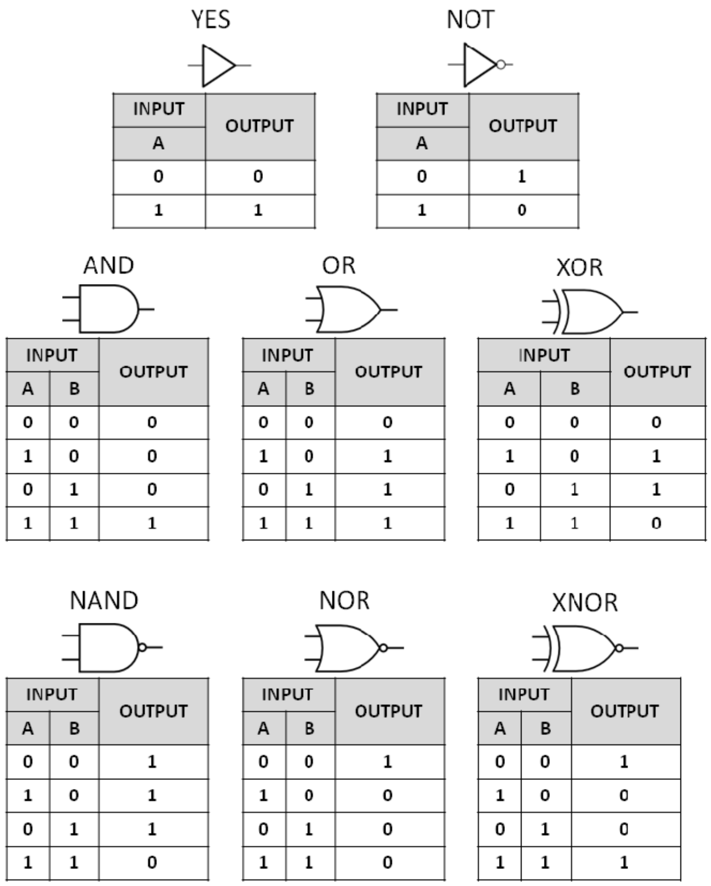
   A. $x \geq 2$
   B. Some text about a variable, x, lajkshgldjk
   C. $\forall x, \exists y = 3 + 5 \to p \lor q \iff a \cup b$
   D. Dots: $x_1, x_2, \ldots, x_n$
   E. Fractions: $\frac{x}{y}$
   F. Times: $a \times b$
   G. Text: $a \times b \quad \text{some text}$
   H. Greek letters: $\pi\gamma\alpha\beta$
   I. $\neg q$
   J. Mod: $5 \pmod 2$
   K. $\mathbb{Z}, \mathbb{R}, \mathbb{N}$
   L. $\binom{73+4-1}{4-1}$ text here.
   M. $\overline{G}$
   N. $1 + \cancel{2x} = a + \cancel{2x}$
   O. Sets: $\{x \mid \exists k \in \mathbb{Z} \text{ s.t.} 2k = x\}$
   P. Sets with large brackets:

$$\left\{x \mid \exists k \in \mathbb{Z} \text{ s.t.} k = \frac{x}{2}\right\}$$

   Q. Cases:

$$\text{sgn}(x) = \begin{cases} 1 & x \geq 0 \\ -2 & x < 0 \end{cases}$$

iv. Using Images



YES

INPUT A	OUTPUT
0	0
1	1

NOT

INPUT A	OUTPUT
0	1
1	0

AND

INPUT A	INPUT B	OUTPUT
0	0	0
1	0	0
0	1	0
1	1	1

OR

INPUT A	INPUT B	OUTPUT
0	0	0
1	0	1
0	1	1
1	1	1

XOR

INPUT A	INPUT B	OUTPUT
0	0	0
1	0	1
0	1	1
1	1	0

NAND

INPUT A	INPUT B	OUTPUT
0	0	1
1	0	1
0	1	1
1	1	0

NOR

INPUT A	INPUT B	OUTPUT
0	0	1
1	0	0
0	1	0
1	1	0

XNOR

INPUT A	INPUT B	OUTPUT
0	0	1
1	0	0
0	1	0
1	1	1

v. Tables

p	q	$p \wedge q$
F	F	F
F	T	F
T	T	T
T	F	F

vi. Proofs

Proof. This is a proof. Having proved it, we can end the proof environment, and a QED symbol will be added for us. □

vii. Skips, Aligns

Centered.

Something Other

$$\forall x, \exists y = 3 + 5 \rightarrow p \wedge q \iff a \cup b$$

Given the equation $x^2 + 5 = 9$, we prove that $x = \pm 2$

$$\begin{aligned} x^2 + 5 &= 9 \\ x^2 &= 4 \\ x &= \pm 2 \end{aligned}$$

It follows that the claim stands.

viii. Text Formatting

(a) Color Box

(b) Text

(c) Text

(d) **Bolded!**

(e) *Text*

(f) More!

ix. Putting it all Together:

Proof. **Given** the equation $x^2 + 5 = 9$, we prove that $x = \pm 2$

$$x^2 + 5 = 9 \tag{1}$$

$$x^2 = 4 \quad \text{Subtract 5 from both sides.} \tag{2}$$

$$x = \pm 2 \quad \text{*(Recall that a sqr gives a pos and neg sol.)} \tag{3}$$

Thus, by mathematically showing that $x = \pm 2$, we have proved our claim. □

x. Comparison

(a) Bad:

We claim the expected value of the maximum number of marshmallows added to one of the three cups of hot chocolate is 4.2. Recall from recitation for finding expected value is (where Pr(s) is the probability of s happening, and X(s) is the random variable we expect of (s): $\sum_{s\in S} X(s)Pr(s)$ In the context of this problem: $\sum_{i=1}^{5} X(i)Pr(i)$ We then substitute, given that the random variable is just the amount of the maximum marshmallows, and the probability for any n marshmallow maximum is the cube of the random variable over 5, minus the probability of all the other 3 way occurrences happening (like for i = 3, when each cup has 2 or each cup has 1 marshmallow). Then, we use WolframAlpha to calculate:
$\sum_{i=1}^{5} i((\frac{i}{5})^3 - (\frac{i-1}{5})^3) = 4.2$
Thus, we have proved our claim.

(b) Good:

Claim: the expected value of the maximum number of marshmallows added to one of the three cups of hot chocolate is 4.2.

Proof. Recall from recitation for finding expected value is (where $Pr(s)$ is the probability of s happening, and $X(s)$ is the random variable we expect of (s):

$$\sum_{s \in S} X(s)Pr(s)$$

In the context of this problem:

$$\sum_{i=1}^{5} X(i)Pr(i)$$

We then substitute, given that the random variable is just the amount of the maximum marshmallows, and the probability for any n marshmallow maximum is the cube of the random variable over 5, minus the probability of all the other 3 way occurrences happening (like for $i = 3$, when each cup has 2 or each cup has 1 marshmallow). Then, we use WolframAlpha to calculate:

$$\sum_{i=1}^{5} i((\frac{i}{5})^3 - (\frac{i-1}{5})^3) = 4.2$$

Thus, we have proved our claim. $\square$

What differences do we notice?